

Number System

- Number is a Math value used for counting or measuring or labelling of.
- Number System is defined as a system of writing to express numbers. It is one the mathematical notation for representing no. of a given set by using digits.

Following are some N.S:- i) Decimal N.S.

ii) Binary N.S.

iii) Octal N.S.

iv) Hexadecimal N.S.

I) Decimal Number System- No. used in day-to-day life.

B

Base = 10 $\Rightarrow$ (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

$$\begin{array}{rcl} \text{eg- } 2586 & = & (2 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (6 \times 10^0) \\ \begin{array}{c} \downarrow \quad \downarrow \quad \downarrow \\ 10^3 \quad 10^2 \quad 10^1 \quad 10^0 \\ \downarrow \quad \downarrow \\ 10^2 \quad 10^0 \end{array} & = & 2000 + 500 + 80 + 6 \\ & & 2586 \end{array}$$

II) Binary N.S.- widely used in Computers. All inputs given to a comp. are 1st converted to B.N.S. b/c it understand only B.N.S..

Base = 2 $\Rightarrow$ (0, 1)

$$\begin{array}{rcl} \text{eg- } 10101 & = & (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ \begin{array}{c} \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \end{array} & = & 16 + 0 + 4 + 0 + 1 \\ & = & (21)_{10} = (10101)_2 \end{array}$$

$$\begin{aligned} (10100011)_2 &= 128 + 0 + 32 + 0 + 0 + 0 + 2 + 1 \\ &= (163)_{10} \end{aligned}$$

III) Octal N.S. - Base $\Rightarrow 8$ (0, 1, 2, 3, 4, 5, 6, 7)

$$\text{eg: } (2057)_8 = 1024 + 0 + 40 + 7$$

$$\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 8^3 & 8^2 & 8^1 & 8^0 \\ 512 & 64 & 8 & 1 \end{array} = (1071)_{10}$$

$$\text{eg: } (2671)_8 = 1024 + 384 + 56 + 1$$

$$= (1465)_{10}$$

$$\text{eg: } (286)_8 = 128 + 64 + 6$$

$$\begin{array}{c} \uparrow \\ \text{invalid} \end{array} = (198)_{10} \times$$

IV) Hexadecimal N.S. - Base $\Rightarrow 16$ (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

16 Symbols or Character (Single)

10 - A	13 - D
11 - B	14 - E
12 - C	15 - F

$$(1AF)_{16} = 256 + 160 + 15$$

$$= (431)_{10}$$

$$(7CF)_{16} = 1792 + 192 + 15$$

$$= (1999)_{10}$$

2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}
1	2	4	8	16	32	64	128	256	512	1024

2^{11}	2^{12}	8^0	8^1	8^2	8^3	8^4
2048	4096	1	8	64	512	4096

16^0	16^1	16^2	16^3	16^4
1	16	256	4096	65536



Converting from Decimal to Binary:-

$$\begin{array}{r}
 (25)_{10} \Rightarrow \begin{array}{r} 2 \ 25 \ 1 \\ 2 \ 12 \ 0 \\ 2 \ 6 \ 0 \\ 2 \ 3 \ 1 \\ 1 \end{array} \Rightarrow (11001)_2 \Rightarrow 16 + 8 + 0 + 0 + 1 \\
 = (25)_{10}
 \end{array}$$

$$\begin{array}{r}
 (42)_{10} \Rightarrow \begin{array}{r} 2 \ 42 \ 0 \\ 2 \ 21 \ 1 \\ 2 \ 10 \ 0 \\ 2 \ 5 \ 1 \\ 2 \ 2 \ 0 \\ 1 \end{array} \Rightarrow (101010)_2
 \end{array}$$

$$\begin{array}{r}
 (156)_{10} \Rightarrow \begin{array}{r} 2 \ 156 \ 0 \\ 2 \ 78 \ 0 \\ 2 \ 39 \ 1 \\ 2 \ 19 \ 1 \\ 2 \ 9 \ 1 \\ 2 \ 4 \ 0 \\ 2 \ 2 \ 0 \\ 1 \end{array} \Rightarrow (10011100)_2
 \end{array}$$

Decimal to Octal:-

$$\begin{array}{r}
 (952)_{10} = \begin{array}{r} 8 \ 952 \ 0 \\ 8 \ 119 \ 7 \\ 8 \ 14 \ 6 \\ 1 \end{array} \Rightarrow (1670)_8
 \end{array}$$

$$\begin{array}{r}
 (210)_{10} = \begin{array}{r} 8 \ 210 \ 2 \\ 8 \ 26 \ 2 \\ 3 \end{array} \Rightarrow (322)_3
 \end{array}$$

Converting Decimal to Hexadecimal:-

$$(428)_{10} = \begin{array}{r} 16 \ 428 \ 12 \\ 16 \ 26 \ 10 \\ 1 \end{array} \Rightarrow (1AC)_{16}$$

$$(188)_{10} = \begin{array}{r} 16 \ 188 \ 12 \\ 11 \end{array} \Rightarrow (BC)_{16}$$

Converting from a base other than 10 to
to base other than 10.

- 1) Convert the original no. to decimal no.
- 2) Convert the decimal no. to req no. (with new base).

$$\text{eg- } (101110)_2 = ()_8 \Rightarrow 32 + 8 + 4 + 2 \\ \Rightarrow (46)_{10}$$

$$\begin{array}{r} 8 \ 46 \ 6 \\ 5 \end{array} \Rightarrow (56)_8$$

$$(1AB)_{16} = 256 + 160 + 11 \\ = (427)_{10}$$

$$(11010011)_2 = 128 + 64 + 16 + 2 + 1 \\ = (211)_{10}$$

$$\begin{array}{r} 2 \ 427 \ 1 \\ 2 \ 213 \ 1 \\ 2 \ 106 \ 0 \\ 2 \ 53 \ 1 \\ 2 \ 26 \ 0 \\ 2 \ 13 \ 1 \\ 2 \ 6 \ 0 \\ 2 \ 3 \ 1 \\ 1 \end{array} \Rightarrow (110101011)_2$$

$$16 \ 211 \ 3 \Rightarrow (D3)_{16}$$

$$2 \ 53 \ 1$$

$$2 \ 26 \ 0$$

$$2 \ 13 \ 1$$

$$2 \ 6 \ 0$$

$$2 \ 3 \ 1$$

$$1$$